

RAMSTEIN, Germany

WMO: 106140

Lat: 49.437N

Lon: 7.600E

Elev: 781

StdP: 14.29

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	13.7	19.1	9.8	9.3	15.5	14.3	11.7	21.3	25.1	46.6	22.2	46.6	3.7	90	0.532

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.3	88.7	68.5	84.5	67.0	80.9	65.7	71.0	82.9	69.0	80.7	67.3	77.9	6.9	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
67.8	105.3	74.4	65.4	96.6	72.0	63.9	91.6	71.0	35.3	83.6	33.5	81.2	32.2	78.2	84.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 19.5	17.1	15.2	DB	6.0	94.6	6.4	3.5	1.4	97.1	-2.3	99.1	-5.9	101.1	-10.5	103.6	(4)
(5)			WB	5.6	74.5	6.3	3.4	1.1	76.9	-2.6	78.9	-6.1	80.8	-10.7	83.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	49.9	34.6	36.5	41.9	48.8	56.0	62.3	66.0	65.0	57.3	50.3	42.1	36.5		
	DBStd	12.95	8.45	7.66	6.86	6.60	6.35	6.33	5.81	5.79	5.67	6.72	7.01	8.07		
	HDD50	1997	478	380	261	100	18	1	0	0	7	81	249	422		
	HDD65	5751	942	799	715	485	286	126	59	72	239	456	688	884		
	CDD50	1942	1	1	10	65	205	370	496	464	225	90	12	3		
	CDD65	223	0	0	0	0	8	45	90	71	8	1	0	0		
	CDH74	3091	0	0	0	31	176	632	1156	941	149	6	0	0		
	CDH80	1022	0	0	0	1	33	186	423	350	29	0	0	0		
	WSAvg	5.9	7.1	6.7	6.7	5.8	5.7	5.3	5.4	4.8	4.9	5.3	5.7	6.9		
	PrecAvg	34.6	3.1	3.0	2.8	2.1	2.7	2.6	2.7	2.8	2.7	3.3	3.1	3.7		
(15) Precipitation	PrecMax	45.7	7.4	7.4	7.2	5.0	5.2	6.0	7.2	7.4	4.8	9.1	6.7	10.7		
	PrecMin	24.1	0.5	0.4	0.5	0.1	0.5	0.9	1.3	0.2	0.8	0.7	0.9	2.0		
	PrecStd	5.9	1.7	1.8	1.7	1.2	1.3	1.3	1.3	1.6	1.2	1.8	1.5	2.1		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	54.0	58.7	68.1	78.4	84.1	90.0	93.6	95.3	84.1	73.8	61.4	55.6		
		MCWB	49.8	49.5	53.2	58.9	65.1	69.8	69.7	69.7	67.0	65.5	54.9	50.7		
	2%	DB	51.9	53.7	62.9	73.2	78.7	85.5	89.3	88.0	78.5	68.3	57.5	53.3		
		MCWB	48.8	47.3	50.9	57.6	63.2	67.5	68.7	68.3	65.6	60.9	53.1	50.0		
	5%	DB	48.6	50.4	58.3	68.1	74.9	80.9	84.5	83.3	73.5	64.5	55.2	50.4		
		MCWB	45.8	46.2	49.4	55.3	61.0	66.0	66.8	66.8	63.1	58.5	51.8	47.8		
	10%	DB	46.4	48.1	53.8	63.9	70.4	77.1	80.6	79.0	69.5	61.1	52.1	48.2		
		MCWB	44.1	45.0	48.2	52.9	58.8	64.0	65.6	65.5	61.0	56.6	49.1	46.3		
	0.4%	WB	51.9	51.6	55.8	62.1	68.3	73.4	73.2	74.6	71.7	67.1	57.4	53.0		
		MCDB	53.5	55.4	63.5	73.3	79.4	85.6	87.2	81.5	78.8	70.2	59.6	54.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	49.3	49.0	53.0	58.4	65.1	69.5	70.4	70.5	66.5	62.1	54.2	50.5		
		MCDB	51.6	52.0	59.5	70.4	75.3	80.0	84.2	82.8	74.6	66.8	56.9	52.3		
	5%	WB	46.4	46.9	50.8	55.8	62.3	67.3	68.5	68.4	64.1	59.3	51.8	48.4		
		MCDB	48.7	49.9	56.0	65.5	71.8	77.9	80.6	80.4	72.1	63.5	54.9	50.2		
	10%	WB	43.8	44.8	48.9	53.8	60.1	65.3	66.9	66.7	61.8	57.1	49.7	46.1		
		MCDB	45.8	47.5	53.7	62.4	68.1	75.1	77.6	76.9	68.3	60.7	52.1	48.1		
(35) Mean Daily Temperature Range	5% DB	MDBR	10.1	13.5	18.1	22.0	21.6	22.0	22.3	22.4	21.1	17.3	11.0	9.3		
		MCDBR	11.0	18.9	28.3	32.4	30.5	29.7	31.5	31.6	29.1	22.8	14.7	11.2		
		MCWBR	9.0	13.8	17.8	17.8	15.4	14.0	12.9	13.8	16.5	15.7	10.8	9.2		
		MCDBR	10.4	16.0	22.1	27.7	26.4	26.5	27.0	27.4	25.5	21.2	13.4	10.3		
	5% WB	MCWBR	9.1	12.7	15.2	16.0	14.1	13.5	12.5	13.7	15.6	15.7	10.6	9.1		
(40) Clear-Sky Solar Irradiance	taub		0.318	0.346	0.378	0.406	0.414	0.417	0.415	0.411	0.384	0.369	0.340	0.316		
	taud		2.398	2.343	2.270	2.232	2.240	2.263	2.259	2.299	2.348	2.378	2.395	2.395		
	Ebn at Noon		231	251	261	266	268	268	266	262	257	239	219	214		
	Edh at Noon		21	28	35	41	42	41	41	37	32	27	21	18		
(44) All-Sky Solar Radiation	RadAvg		260	495	874	1361	1610	1799	1704	1460	1097	633	294	205		
	RadStd		39	91	133	175	170	156	173	150	130	59	31	30		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (42 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page